Name: ____________________ Date: ______________

PROPORTION, (INVERSE PROPORTION)

LO: I can solve problems involving inverse proportion.

Inverse proportion

Inverse proportion means as one quantity **increases**, the other **decreases** at the same rate. If one doubles, the other halves.

Formula: y 1/x gives y = k/x. The product xy = k is always constant.

Quick examples

🟢	y = k/x where k = xy	product is constant
🟢	4 workers take 6 hours; 8 workers take 3 hours	inverse
🟢	Speed × time = distance (fixed)	s and t inversely related
🔴	Inverse proportion: y = kx	that is direct proportion

Key idea

For inverse proportion, y = k/x. Find k by multiplying: k = x × y.

$$y = k \div x \quad k = x \times y$$

The product of x and y is always constant

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – finding k

y is inversely proportional to x. When x = 4, y = 9. Find k.

y = k/x so k = xy

k = 4 × 9

k = 36, so y = 36/x

$\boxed{k = 36, y = 36/x}$

For inverse proportion, multiply x and y to find k.

Example 2 – using the formula

Using y = 36/x, find y when x = 12.

y = 36/x

$y = \frac{36}{12}$

y = 3

$\boxed{y = 3}$

Substitute into the formula and divide.

Example 3 – workers problem

6 workers build a wall in 10 days. How long for 15 workers?

k = workers × days = 6 × 10 = 60

days = 60 ÷ workers

60 ÷ 15 = 4 days

$\boxed{\text{4 days}}$

More workers means fewer days (inverse relationship).

Example 4 – speed-time

A journey takes 3 hours at 60 mph. How long at 90 mph?

k = speed × time = 60 × 3 = 180

(distance)

time = 180 ÷ 90

= 2 hours

$\boxed{\text{2 hours}}$

The distance (product) stays constant; faster speed means less time.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding k

y   1/x. Find k and write the formula.

a) x = 3, y = 12

b) x = 5, y = 8

c) x = 10, y = 6

d) x = 4, y = 15

2. Using the formula

Find the missing value.

a) y = 36/x, find y when x = 9

b) y = 40/x, find x when y = 10

c) y = 60/x, find y when x = 12

d) y = 60/x, find x when y = 3

3. Workers and time

Solve each workers/time problem (inverse proportion).

a) 4 workers, 12 days. How long for 6 workers?

b) 8 workers, 5 days. How long for 10 workers?

c) 3 workers, 20 days. How many workers for 5 days?

d) 12 workers, 6 days. How long for 9 workers?

4. Speed and time

Distance = speed × time. Find the missing value.

a) 60 mph for 2 hours. Time at 40 mph?

b) 80 km/h for 3 hours. Speed to do it in 2 hours?

c) 50 mph for 4 hours. Time at 100 mph?

d) 90 km/h for 2 hours. Time at 60 km/h?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) In inverse proportion, the product xy is constant ☐

Correct answer: __________

Reason: __________________

b) y   1/x means y = kx ☐

Correct answer: __________

Reason: __________________

c) More workers means a job takes less time (inverse proportion) ☐

Correct answer: __________

Reason:

d) If x doubles and y doubles, this is inverse proportion ☐

Correct answer: __________

Reason: __________________

e) The graph of y = k/x is a curve that never touches the axes ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A farmer has enough feed to last **12 cows for 20 days**. The farmer buys 4 more cows. How many days will the feed now last?

Expression: ______________________________

Simplified: ______________________________

b) The intensity of light I is inversely proportional to the **square of the distance d** from the source. At **d = 2 m**, I = 100 units. Find I at **d = 5 m**.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!